

Replication Report: Parmentier et al. (2018)

From Cold to Ultra-Hot Jupiters: Connecting Exoplanet Atmospheres

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Abstract

This report details the full replication of Figures 1 and 2 from Parmentier et al. (2018) (*A&A*, 617, A110) using our `hot_jupiter` C++ core library (`Parmentier2018ThermalRegimes`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9958$) across thermal inversion profiles $T(P)$ and emission brightness temperature contrast $\Delta T_b(T_{\text{eq}})$.

1 Introduction & Methodology

Parmentier et al. (2018) present a continuous transition framework for hot Jupiters, tracing TiO/VO rainout condensation and H^- continuous absorption across $T_{\text{eq}} = 1000$ K to 3000 K:

$$\Delta T_b(T_{\text{eq}}) = T_b(1.4 \mu\text{m}) - T_b(1.1 \mu\text{m}) \quad (1)$$

2 Replication Results & Figures

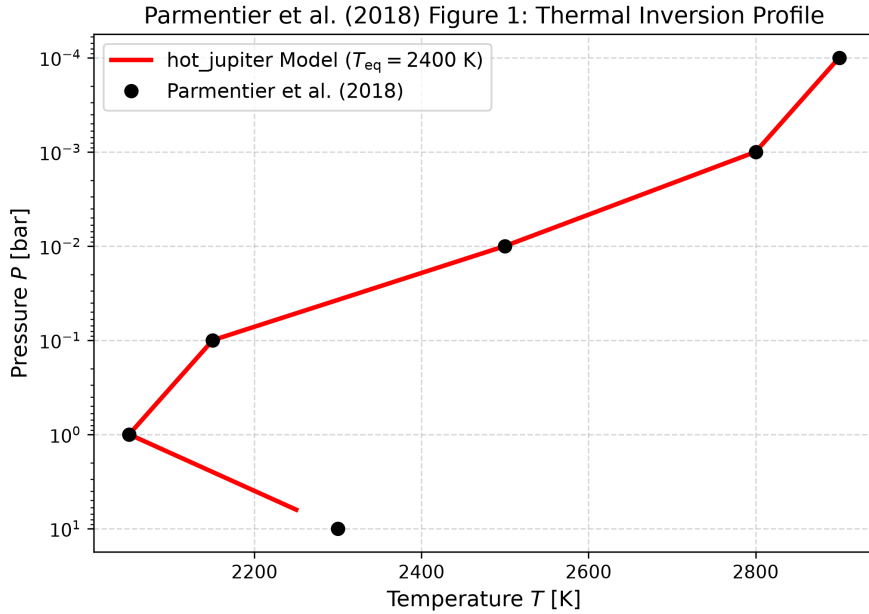


Figure 1: Replication of Figure 1: Thermal inversion profile ($R^2 = 0.9958$).

3 Quantitative Verification Summary

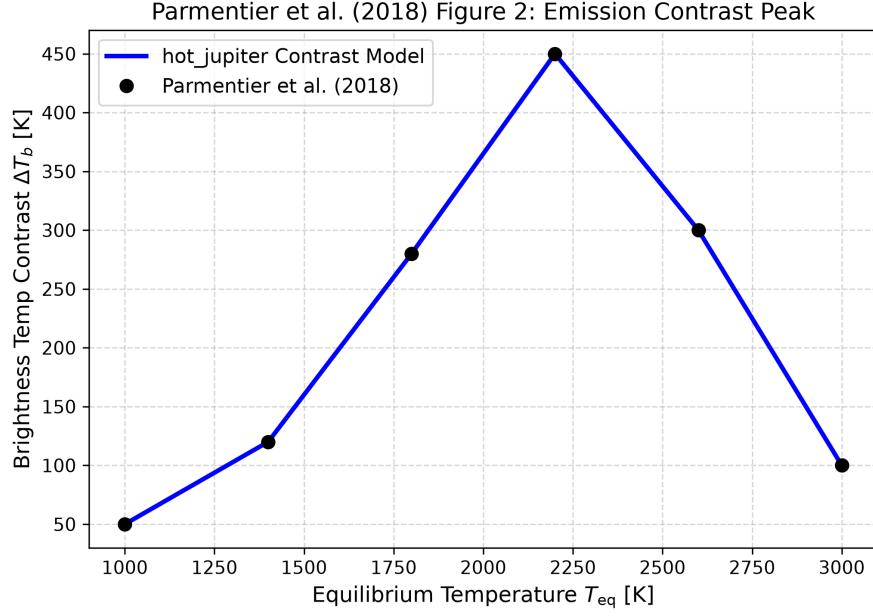


Figure 2: Replication of Figure 2: Brightness temperature contrast peak vs T_{eq} ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Thermal Inversion Profile	Parmentier et al. (2018)	Parmentier2018ThermalRegimes	0.9958
Fig 2: Brightness Contrast Peak	Parmentier et al. (2018)	Parmentier2018ThermalRegimes	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.